

Database Systems

Assignment03



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Q1:

1. List the names of countries that have reported critical-level threats (Severity_Level = "Critical") in 2025:

$$\pi \text{ Country_Name } ((\sigma \text{ Severity_Level} = \text{"Critical"} \wedge \text{Year}(\text{Date_Report}) = 2025 \\ (\text{THREAT} \bowtie \text{Threat_ID} = \text{Threat_ID} \text{ INCIDENT})) \bowtie \text{Country_ID} = \text{Country_ID} \text{ COUNTRY})$$

Explanation:

Select incidents from 2025 with Critical severity, join with corresponding threats and countries, and project country names.

2. Retrieve the names of response teams that successfully resolved ransomware-related incidents:

$$\pi \text{ Team_Name } ((\sigma \text{ Threat_Name} = \text{"Ransomware"} \wedge \text{Resolution_Status} = \text{"Resolved"}) \\ (\text{THREAT} \bowtie \text{Threat_ID} = \text{Threat_ID} (\text{INCIDENT} \bowtie \text{Incident_ID} = \text{Incident_ID} \\ \text{ACTION_TAKEN} \bowtie \text{Team_ID} = \text{Team_ID} \text{ RESPONSE_TEAM})))$$

Explanation:

Join all related tables, filter ransomware threats that were resolved, and project the response team names.

3. Find all countries that have not yet formed any cyber response team:

$$\pi \text{ Country_Name } (\text{COUNTRY}) - \pi \text{ Country_Name } (\text{COUNTRY} \bowtie \text{Country_ID} = \text{Country_ID} \\ \text{RESPONSE_TEAM})$$

Explanation:

Subtract countries that have response teams from the list of all countries to find those without one.

4. Display the names of threats that have been reported by more than one country:
$$\pi \text{ Threat_Name } (\sigma \text{ count(Country_ID)} > 1 (\gamma \text{ Threat_ID,} \\ \text{count(Country_ID)} \rightarrow \text{countCountry (INCIDENT)} \bowtie \text{Threat_ID = Threat_ID THREAT}))$$
Explanation:

Group incidents by threat, count distinct reporting countries, and return threats reported by more than one country.

5. List all AI-based attack incidents (Category = "AI Attack") along with their country names and impact scores greater than 80:
$$\pi \text{ Threat_Name, Country_Name, Impact_Score } (\sigma \text{ Category = "AI Attack" } \wedge \\ \text{Impact_Score} > 80 (\text{THREAT} \bowtie \text{Threat_ID = Threat_ID (INCIDENT} \bowtie \text{Country_ID =} \\ \text{Country_ID COUNTRY))})$$
Explanation:

Select AI Attack incidents with an impact score above 80, join related tables, and display the threat name, country, and impact score.

Q2:

1. Relational Algebra:

$\pi_{\text{itemName}} (\sigma_{\text{isAvailable} = 0 \wedge \text{price} > 500} (\text{MenuItem}))$

Textual Meaning:

Selects the names of menu items that are currently unavailable and have a price greater than 500.

SQL Query:

```
SELECT itemName  
FROM MenuItem  
WHERE isAvailable = 0 AND price > 500;
```

2. Relational Algebra:

$\pi_{\text{itemName}} (\sigma_{(\text{category} = \text{'Beverage'} \wedge \text{price} > 300) \vee \text{category} = \text{'Dessert'}} (\text{MenuItem}))$

Textual Meaning:

Retrieves the names of menu items that are either beverages priced above 300 or belong to the dessert category.

SQL Query:

```
SELECT itemName  
FROM MenuItem  
WHERE (category = 'Beverage' AND price > 300) OR category = 'Dessert';
```

3. Relational Algebra:

$\pi_{\text{itemName}, \text{price}} (\sigma_{\text{category} = \text{'Snack'} \wedge \text{isAvailable} = 1} (\text{MenuItem}))$

Textual Meaning:

Displays the names and prices of snack items that are available.

SQL Query:

```
SELECT itemName, price
FROM MenuItem
WHERE category = 'Snack' AND isAvailable = 1;
```

4. Relational Algebra:

$\pi_{\text{itemName}, \text{price}/100} (\sigma_{\text{category} = \text{'Snack'} \wedge \text{isAvailable} = 1} (\text{MenuItem}))$

Textual Meaning:

Shows the names of available snack items along with their price divided by 100.

SQL Query:

```
SELECT itemName, price/100 AS price_in_hundreds
FROM MenuItem
WHERE category = 'Snack' AND isAvailable = 1;
```

5. Relational Algebra:

$\pi_{\text{supplierID}} (\text{Supplies} \bowtie \text{MenuItem} \bowtie \sigma_{\text{itemName} = \text{'Cappuccino'}} (\text{MenuItem}))$

Textual Meaning:

Finds the IDs of suppliers who have supplied the item named “Cappuccino”.

SQL Query:

```
SELECT DISTINCT supplierID
FROM Supplies
JOIN MenuItem ON Supplies.itemID = MenuItem.itemID
WHERE itemName = 'Cappuccino';
```

6. Relational Algebra:

$\pi_{\text{Employee.name, Cafe.cafeName}} (\sigma_{\text{Employee.cafeID} = \text{Cafe.cafeID} \wedge \text{Employee.address} = \text{Cafe.city}} (\text{Employee} \bowtie \text{Cafe}))$

Textual Meaning:

Retrieves the names of employees and their café names where the employee works in a café located in the same city as their address.

SQL Query:

```
SELECT Employee.name, Cafe.cafeName
FROM Employee
```

JOIN Cafe ON Employee.cafeID = Cafe.cafeID
WHERE Employee.address = Cafe.city;

7. Relational Algebra:

$\pi_{\text{empID}} (\sigma_{\text{salary} \neq 50000} (\text{Employee}))$

Textual Meaning:

Lists the IDs of employees whose salary is not equal to 50000.

SQL Query:

SELECT empID
FROM Employee
WHERE salary <> 50000;

8. Relational Algebra:

$\pi_{\text{supplierID}} (\sigma_{\text{city} = \text{'Karachi'}} (\text{Supplier})) \cap \pi_{\text{supplierID}} (\sigma_{\text{city} = \text{'Lahore'}} (\text{Supplier}))$

Textual Meaning:

Finds supplier IDs of those who operate in both Karachi and Lahore.

SQL Query:

```
SELECT supplierID  
FROM Supplier  
WHERE city = 'Karachi'  
INTERSECT  
SELECT supplierID  
FROM Supplier  
WHERE city = 'Lahore';
```

9. Relational Algebra:
$$\pi_{\text{itemName}} (\text{MenuItem}) - \pi_{\text{itemName}} (\text{MenuItem} \bowtie \text{Supplies})$$
Textual Meaning:

Retrieves the names of menu items that have never been supplied by any supplier.

SQL Query:

```
SELECT itemName  
FROM MenuItem  
WHERE itemID NOT IN (SELECT itemID FROM Supplies);
```